

The empirical recurrence for OEIS A236091, recovered from its tail

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Abstract

OEIS A236091 records “Empirical recurrence of order 73” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 1512$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 6$, so the recurrence is proved for every $n \geq 79$. Its last coefficient is $c_{73} = 51200 \neq 0$, so no further tail satisfies anything shorter; any order-73 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A236091 is “Number of $(n+1) \times (3+1) \times 0.2$ arrays colored with the difference of the maximum and the upper median in each 2×2 subblock.”. It has offset 1 and begins

1512, 10912, 78302, 495192, 3614408, 26325004, 206365422, 1454459216, ...

The entry states:

Empirical recurrence of order 73 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 09:05:40, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 1512$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 1512$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 73: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 6$ is the first whose minimal order is exactly the stated 73.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 3024$ exact terms from $n = 6$ on, it returns order 73 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{73} c_i a(n-i),$$

c_1	0	c_{20}	88786773870	c_{39}	109505101121016	c_{58}	−1527091686016
c_2	0	c_{21}	9738922000	c_{40}	148142789856552	c_{59}	2181932640640
c_3	0	c_{22}	1009279198040	c_{41}	−157115300761072	c_{60}	−984483107136
c_4	2854	c_{23}	97351255698	c_{42}	9014602365068	c_{61}	235927164416
c_5	148	c_{24}	−773062848880	c_{43}	−68012438906584	c_{62}	−155310500480
c_6	28898	c_{25}	−453544002852	c_{44}	−265667443273184	c_{63}	13570499072
c_7	5084	c_{26}	−9163116878455	c_{45}	247830171808000	c_{64}	13229167744
c_8	−876701	c_{27}	−2205871843964	c_{46}	8654932139628	c_{65}	−6218326272
c_9	−1784	c_{28}	4013713021515	c_{47}	−107776559421936	c_{66}	3889533952
c_{10}	−9405193	c_{29}	5159934301310	c_{48}	184708521842848	c_{67}	−363442176
c_{11}	−1091496	c_{30}	43883043243201	c_{49}	−146121843386384	c_{68}	−74889472
c_{12}	97829615	c_{31}	14293725461260	c_{50}	−19868986379936	c_{69}	12107264
c_{13}	−5186866	c_{32}	−2614483232228	c_{51}	99926174920864	c_{70}	−9734912
c_{14}	1065140089	c_{33}	−31763282419308	c_{52}	−13125192546448	c_{71}	−2049024
c_{15}	60235820	c_{34}	−100203636889000	c_{53}	−46466514753472	c_{72}	−215040
c_{16}	−4495630350	c_{35}	−47302265149110	c_{54}	31800434790096	c_{73}	51200
c_{17}	314875004	c_{36}	−46572482466620	c_{55}	−12294930683328		
c_{18}	−49983823705	c_{37}	102332468093308	c_{56}	−2732363201952		
c_{19}	−1272802386	c_{38}	67433892752277	c_{57}	3862872849472		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 79$, and it is the recurrence OEIS A236091 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 6$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{73} - c_1 x^{72} - \dots - c_{73}$. Its constant term is $-c_{73} = -51200$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of

terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 73 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 73, so $\deg q = 73$, and it is monic. It holds from some index t on. From $\max(t, 6)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 73, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 17 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 73, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 51200.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-73 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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